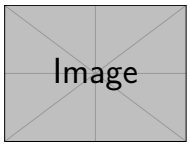
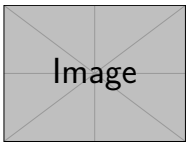
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A [multilateral/multi-objective] perspective towards [supply chain network design] in an uncertain environment: Methodology and implementation

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ABSTRACT

Motivated by a real-life case study, this paper proposes a [multilateral perspective / novel framework] for [supply chain network / domain] design as an intricate decision-making problem. The proposed approach utilizes a novel [multi-objective mathematical model] by incorporating both quantitative and qualitative factors to comprehensively model the investigated case study. In the proposed model, the quantitative factors aim to [minimize / maximize] [objective 1] and [objective 2]. Moreover, the number of [facilities], which are capable of [performing function], are [maximized / minimized] through a formulation based upon qualitative criteria. Moreover, to cope with the uncertainty, a robust optimization approach is utilized, and an interactive fuzzy solution approach is applied to solve the proposed multi-objective model. The computational results demonstrate the capability of the proposed model in designing the [network] in practice. Moreover, some sensitivity analyses are conducted to show the findings observed in a real case study. The results show that the decision makers should be aware of [quality-based / cost-based] requirements, because of significant impact on [facility location / resource allocation], besides the [cost / quality]-oriented consideration. Finally, based on the finding, helpful managerial insights are provided for the decision makers in the [domain] network design.

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1. Introduction

The world fast-growing population has resulted in an increase in [diseases / costs / demand] which redouble the importance of [domain] considerations. Among the items used, [product] and its sub-products are crucial commodities in any [sector] because they cannot be replaced by any alternatives, and each of them has its special usage. In addition, [domain]-related cost owns a significant part of total [sector] expenditures. By 2040, [relevant projection] expenditure is expected to have an approximate growth of [X]% [**?**]. Therefore, [domain] supply chain (DSC) management due to its inherent importance in [sector] has become an important decision-making problem and drawn attention from researchers.

[Domain] supply chains are complex and difficult to manage owing to the fact that both supply and demand are tainted with uncertainty. Since [source] is the only source of [product] with almost unpredictable behaviors, [product] supply is an unavoidably uncertain parameter. According to [Organization], on average, only [X]% of potential [suppliers] supplied [product] in [year] [**?**].

Generally, uncertain environments can be defined in three types namely stochastic, fuzzy and robust each of which are dealt with their corresponding modeling approaches [**?**]. It is noteworthy that the stochastic programming approach comes to the application when the adequate information about distribution functions of random data is available. However, the robust approach is applicable in the situation of little knowledge about the historical data [**?**]. Consequently, it is of particular importance to make robust decisions, especially for issues with a long-term horizon.

In this relation, this study sets out to give proper re-

sponses to the following main questions:

1. What is the influential qualitative and quantitative factors that affect the decisions in [domain] network design? Which approach is appropriate to cope with? And how can make a tradeoff these factors to effective and efficient performance for [domain]?
2. Which aspects of [domain] are tainted by uncertainty? How to formulate the uncertainty in the proposed model? And what are the impacts of uncertainty on the proposed model?
3. How to embed specific characteristics of [product] like a shelf life time, different [policies], and optimal inventory level of [product] in the proposed model?

In this paper, we utilize effective multi-attribute group decision-making (MAGDM) approach to account for a variety of qualitative attributes like [attribute 1], [attribute 2], and [attribute 3] which are evaluated under the preferences of a group of beneficiaries as the decision makers (DMs). A three-phase multi-attribute multi-objective robust (3P-MAMOR) approach is also devised to cope with the uncertain environment, and then, an interactive fuzzy approach is provided to handle the objectives multiplicity [**? ?**].

The rest of the paper is organized as follows. Section 2 reviews the related literature. In Section 3, the solution methodology is described, and the problem description and mathematical formulation are provided in Section 4. Section 5 describes the case study in more details, and Section 6 outlines the computational experiments. Finally, Section 7 provides discussions, concluding remarks and future research directions.

2. Literature Review

In this section, an extant body of literature is investigated in the form of a structured review. According to the review by **?**], the studies were conducted based on the issues defined on hierarchical levels. Another survey was provided by **?**] classifying the papers based on the echelons of the supply chain. According to Table **??**, the papers are categorized based on hierarchical level, planning horizon, mathematical modelling approach, time periods, solution technique, type of [product], objective function multiplicity, performance measure, and uncertainty modelling approach.

2.1. [Stage 1] in [Domain] Supply Chains

Some papers have considered the issues on [stage 1] level [**? ? ?**]. Different [policies] were evaluated by a simulation model developed by **?**]. Results indicated that [collecting / producing] as much as possible was not necessarily convenient. **?**] addressed [domain] network design while considering social aspects.

2.2. [Stage 2] in [Domain] Supply Chains

A number of works have accounted for [stage 2] planning for [product] [**? ?**]. A decision support system (DSS) for [product] production planning in regional centers was addressed by **?**] through an integer non-linear programming model.

2.3. Uncertainty in [Domain] Supply Chains

To the best of our knowledge, the majority of the studies which have addressed the relevant issues in an uncertain environment and to tackle uncertainty, stochastic programming approach is widely applied [**? ? ?**]. Among them, robust optimization approaches and its derivatives have got scant attention [**? ? ?**]. The aforementioned literature indicates that the inclusion of both quantitative and qualitative attributes while designing a [domain] network robustly performing in an uncertain environment is a critical factor which has not got apposite attention. To fill this gap, this paper proposes a realistic approach to investigate the problem from 3P-MAMOR strategic decision-making perspective which is in line with the development of previous studies in the literature by addressing the following main contributions:

- A novel multi-objective model for the simultaneous consideration of [objective 1] (by investigating quantitative attributes), [objective 2], and most favorable locations for [facilities] (based on qualitative factors evaluated by MAGDM approach);
- A realistic multi-product multi-period [domain] network by considering three key issues (the design and location of facilities, perishability of [products], and uncertainty in critical parameters);
- A three-phase multi-attribute multi-objective robust approach to cope with a group decision-making network design problem in an uncertain environment;
- Using a real case study of [Location] to illustrate the applicability of the proposed model and solution.

3. Methodology

In this section, we suggest a 3P-MAMOR approach to appropriately design the [domain] supply chain network. The proposed approach accounts for all information about attributes and qualitative factors involving in facility location decisions by an effective MAGDM technique. It also protects the proposed multi-objective model against uncertainties by applying a robust optimization approach and devises an interactive fuzzy approach, called TH [**? ?**], to solve the model.

The schematic of the proposed 3P-MAMOR framework can be demonstrated in Fig. **??**. In phase 1 (identification), most appropriate locations for [facilities] among the candidate ones are identified. In phase 2 (protection), we confront an unavoidable uncertainty of the parameters and protect the solutions. In phase 3 (integration), the cost-oriented

and quality-oriented attitudes are integrated into a single objective structure.

3.1. Phase 1: The Multi-Attribute Group Decision-Making Approach

Decisions on the locations of facilities must be made with respect to some critical factors which can be [type] considerations. Indeed, we have to deal with MADM environment in which each qualitative attribute is evaluated by employing MAGDM perspective based on the preferences of a group of DMs, each of which differentiates in giving weights to the attributes.

*3.1.1. Method Framework.*The value of alternative A_i on criterion C_j is shown by the interval-valued intuitionistic fuzzy setting (IVIFS) where $A_i = \{\langle C_j, [\mu_{A_iL}(C_j), \mu_{A_iU}(C_j)], [\vartheta_{A_iL}(C_j), \vartheta_{A_iU}(C_j)]\rangle \mid C_j \in C\}$, such that $0 \le \mu_{A_iU}(C_j) + \vartheta_{A_iU}(C_j) \le 1$, $\mu_{A_iL}(C_j) \ge 0$, $\vartheta_{A_iL}(C_j) \ge 0$, $i = 1,2,\ldots,m$ and $j = 1,2,\ldots,n$. The IVIFS is defined by $r_{ij} = \langle [a_{ij}, b_{ij}], [c_{ij}, d_{ij}]\rangle$ which represents a pair of intervals for each $C_j \in C$. Thus, R^k implies on the interval-valued intuitionistic fuzzy decision matrix k ($k = 1,2,\ldots,t$) as follows [**?**]:

$$R^k = \begin{pmatrix} \langle [a^k_{11}, b^k_{11}], [c^k_{11}, d^k_{11}]\rangle & \cdots & \langle [a^k_{1n}, b^k_{1n}], [c^k_{1n}, d^k_{1n}]\rangle \\ \vdots & \ddots & \vdots \\ \langle [a^k_{m1}, b^k_{m1}], [c^k_{m1}, d^k_{m1}]\rangle & \cdots & \langle [a^k_{mn}, b^k_{mn}], [c^k_{mn}, d^k_{mn}]\rangle \end{pmatrix} \tag{1}$$

*3.1.2. The Weights of DMs.*Based on matrix R^k introduced above, the score matrix as well as the average score matrix for each DM can also be calculated [**?**]; and the collective correlation coefficient can be defined as:

$$E_k = \sum_{i=1}^{m} \frac{\sum_{j=1}^{n} S^k_{ij} S^*_{ij}}{\sqrt{\sum_{j=1}^{n} (S^k_{ij})^2} \sqrt{\sum_{j=1}^{n} (S^*_{ij})^2}} \tag{2}$$

where S^k_{ij} and S^*_{ij} represent the score matrix and the average score matrix, respectively. Finally, the weight vector $\Gamma = (\Gamma_1, \Gamma_2, \ldots, \Gamma_t)$ for DMs is constructed by:

$$\Gamma_k = \frac{E_k}{\sum_{k=1}^{t} E_k} \tag{3}$$

where $0 \le \Gamma_k \le 1$, $\sum_{k=1}^{t} \Gamma_k = 1$.

3.1.3. Weights of Attributes and Ranking the Alternatives. The interval-valued intuitionistic fuzzy decision-making problems can be developed by means of integrated programming models [**?**] as follows:

$$\begin{aligned} \text{Max } Q(\omega) &= \sum_{i=1}^{m} \sum_{j=1}^{n} \omega_j S_{ij} \\ \text{s.t.} \quad & \sum_{j=1}^{n} \omega_j = 1, \quad \omega \in I, \quad \omega_j \ge 0 \end{aligned} \tag{4}$$

Finally, the alternatives are analyzed using the simple additive weighting (SAW) method:

$$u(A_i) = \sum_{j=1}^{n} \omega_j S_{ij} \tag{5}$$

where S_{ij} denotes the collective score matrix and $u(A_i)$ denotes the utility of ith alternative.

3.2. Phase 2: Robust Optimization Approach

In this section, a robust optimization approach is described based on **? ?**]. We consider the following MIP model assuming X is a convex polyhedron:

$$\begin{aligned} &\text{Min} \sum_j c_j x_j \\ &\text{s.t.} \sum_j \tilde{a}_{ij} x_j \le b_i \quad \forall i \\ &x \in X \end{aligned} \tag{6}$$

where c represents the n-vector of objective coefficients, and b_i is the m-vector of RHS related to constraint i. The uncertain coefficient $\tilde{a}_{ij}$ takes values in a symmetric and bounded interval represented as $[a_{ij} - \hat{a}_{ij},\ a_{ij} + \hat{a}_{ij}]$ in which $\hat{a}_{ij}$ denotes the worst case. By defining the protection function $\beta(x, \Gamma_i)$ as Eq. (**??**), each uncertain constrain i is rewritten as Eq. (**??**).

$$\beta(x, \Gamma_i) = \max_{\tilde{a}_{ij} \in J_i} \left\{ \sum_j \hat{a}_{ij} x_j \right\} \quad \forall i \tag{7}$$

$$\sum_j a_{ij} x_j + \beta(x, \Gamma_i) \le b_i \quad \forall i \tag{8}$$

Additionally, the robust model can be formulated using LP duality as below:

$$\begin{aligned} &\text{Min} \sum_j c_j x_j \\ &\text{s.t.} \sum_j a_{ij} x_j + Z_i \Gamma_i + \sum_j p_{ij} \le b_i \quad \forall i \\ &Z_i + p_{ij} \ge \hat{a}_{ij} x_j \quad \forall i, j \\ &p_{ij} \ge 0 \quad \forall i, j \\ &Z_i \ge 0 \quad \forall i \\ &x_j \ge 0 \quad \forall j \end{aligned} \tag{9}$$

3.3. Phase 3: The Fuzzy Programming Solution Approach

This paper utilizes an efficient single-phase technique the TH method introduced by **?**] to deal with the concerned MOLP model. Let the compact MOLP model be:

$$\begin{aligned} &\text{Min } Z_1, \quad \text{Min } Z_2, \quad \text{Max } Z_3 \\ &\text{s.t.} \quad Ax \le b, \quad x \in X \end{aligned} \tag{10}$$

Step 1: Calculate the positive and negative ideal solutions (PIS and NIS, respectively) for objective function h;

$h = 1,2,3$.

$$Z_1^{\delta-\text{NIS}} = \text{Max}\{Z_1(x_2^{\delta-\text{PIS}}), Z_1(x_3^{\delta-\text{PIS}})\} \quad (11)$$

$$Z_2^{\delta-\text{NIS}} = \text{Max}\{Z_2(x_1^{\delta-\text{PIS}}), Z_2(x_3^{\delta-\text{PIS}})\} \quad (12)$$

$$Z_3^{\delta-\text{NIS}} = \text{Min}\{Z_3(x_1^{\delta-\text{PIS}}), Z_3(x_2^{\delta-\text{PIS}})\} \quad (13)$$

Step 2: Calculate the satisfaction level of objective function h; $h = 1,2,3$ under solution vector x:

$$\mu_1(x) = \begin{cases} 1 & \text{if } Z_1 < Z_1^{\delta-\text{PIS}} \\ \dfrac{Z_1^{\delta-\text{NIS}} - Z_1}{Z_1^{\delta-\text{NIS}} - Z_1^{\delta-\text{PIS}}} & \text{if } Z_1^{\delta-\text{PIS}} \le Z_1 \le Z_1^{\delta-\text{NIS}} \\ 0 & \text{if } Z_1 > Z_1^{\delta-\text{NIS}} \end{cases} \quad (14)$$

$$\mu_3(x) = \begin{cases} 1 & \text{if } Z_3 > Z_3^{\delta-\text{PIS}} \\ \dfrac{Z_3 - Z_3^{\delta-\text{NIS}}}{Z_3^{\delta-\text{PIS}} - Z_3^{\delta-\text{NIS}}} & \text{if } Z_3^{\delta-\text{PIS}} \le Z_3 \le Z_3^{\delta-\text{NIS}} \\ 0 & \text{if } Z_3 < Z_3^{\delta-\text{NIS}} \end{cases} \quad (15)$$

Step 3: Provide the single objective formulation based on the TH aggregation function:

$$\begin{aligned} \text{Max } & \lambda(x) = \Gamma * \lambda_0 + (1-\Gamma)\sum_h \vartheta_h * \mu_h(x) \\ \text{s.t. } & \lambda_0 \le \mu_h(x), \quad h = 1,2,3 \\ & x \in F(x)\lambda_0,\ \lambda \in [0,1] \\ & \sum_h \vartheta_h = 1, \quad \vartheta_h \ge 0 \end{aligned} \quad (16)$$

4. Problem Statement

The designed [domain] supply chain network in this paper is based on a real case study and it contains [N] echelons including [echelon 1], [echelon 2], [echelon 3], [echelon 4], and [echelon 5], as shown in Fig. **??**.

- **[Echelon 1]:** [Product] supply does not have any particular pattern and is tainted with uncertainty since [source] behavior is not completely known or predictable.
- **[Echelon 2]:** [Collection / production] process is generally covered by mobile and local facilities, which are distinguished by their capacities, functions, and costs.
- **[Echelon 3]:** Being received by regional centers, [product] units are tested to make sure they are safe against [hazards] and capable of being fractionated into subproducts.
- **[Echelon 4]:** The demand for [product] is accompanied with a high level of uncertainty and has no regular order.
- **[Echelon 5]:** The final destination of discarded [product] is a disposal center.

4.1. Notations

The following indices, parameters, and decision variables will be used for the purpose of model formulations (uncertain parameters are represented with a tilde on):

4.1.1. Indices.

I	Index of ith [donor/source] group ($i, i' \in I$).
J_1	Index of locations for j_1th fix [facility] ($j_1 \in J_1$).
J_2	Index of locations for j_2th mobile [facility] ($j_2 \in J_2$).
J	Index of locations for jth [facility] ($j \in J$, $j = j_1 \cup j_2$).
K	Index of locations for kth regional center ($k \in K$).
R	Index of locations for rth disposal center ($r \in R$).
P	Index of pth [product] ($p \in P$).
M	Index of mth [method] ($m \in M$).
H	Index of hth demand zone ($h \in H$).
T	Index of tth time period ($t, t' \in T$).

4.1.2. Binary Variables.

Y_j	1 if jth [facility] j is established; 0 otherwise.
Z'_{jk}	1 if jth [facility] is allocated to kth regional center; 0 otherw
Z_{ij}	1 if ith [source] group is allocated to jth [facility]; 0 otherwi

4.2. Objective Functions

The proposed multi-objective formulation is structured as follows:

4.2.1. First Objective Function. The first objective function seeks to minimize the total network cost involving establishment cost (EC), operating cost (OC), transportation costs (TC), inventory cost (IC), production cost (PC) and unusable cost (UC) during multiple periods:

$$\text{Min } Z_1 = EC + OC + TC + IC + PC + UC \quad (17)$$

$$EC = \sum_{j_1} \left(\tilde{E}_{j_1} + \tilde{F}_{j_1} + \tilde{LA}_{j_1}\right) * Y_{j_1} + \sum_{j_2} \left(\tilde{E}_{j_2} + \tilde{F}_{j_2}\right) * Y_{j_2} \quad (18)$$

$$OC = \sum_m \sum_i \sum_j \sum_t \widetilde{OC}_m * V_{mijt} \quad (19)$$

$$TC = \sum_m \sum_j \sum_k \sum_t \widetilde{TC}_{jk} * V'_{mjkt} + \sum_k \sum_h \sum_p \sum_t \sum_{t'} \widetilde{TC}'_{kh} * W_{khptt'} \quad (20)$$

$$IC = \sum_p \sum_k \sum_t \sum_{t'} \widetilde{IC}_{pkt} * B_{pktt'} + \sum_p \sum_h \sum_t \sum_{t'} \widetilde{IC}'_{pht} * B'_{phtt'} \quad (21)$$

$$PC = \sum_p \sum_m \sum_k \sum_t \widetilde{PC}_{pm} * Q_{pmkt} \quad (22)$$

$$UC = \sum_{p}\sum_{k}\sum_{r}\sum_{t}\widetilde{TD}_{kr} * E_{pkrt} + \sum_{p}\sum_{k}\sum_{r}\sum_{t}\widetilde{TD}'_{kr} * E'_{pkrt'} + \sum_{p}\sum_{h}\sum_{r}\sum_{t}\widetilde{TD}_{hr} * E_{phrt'} \quad (23)$$

*4.2.2. Second Objective Function.*The second objective function utilizes the following linear relation of [product] freshness introduced by **?**]:

$$\text{Min } Z_2 = \sum_{p}\varphi_p \quad (24)$$

$$\varphi_p = \sum_{k}\sum_{h}\sum_{t}\sum_{t'}\vartheta_p * \text{Max}\left\{QLP_p, \frac{t'-t}{L_p}\right\} * \left(B_{pktt'} + B'_{phtt'}\right) \quad (25)$$

where φ_p is quality loss function of pth product, QLP_p represents the point of quality reduction of pth product, and L_p defines product's shelf life.

*4.2.3. Third Objective Function.*Locating [facilities] is an intricate decision making involving both qualitative and quantitative attributes:

$$\text{Max } Z_3 = \sum_{m}\sum_{i}\sum_{j_1}\sum_{t}QF_{j_1} * V_{m,i,j_1,t} + \sum_{m}\sum_{i}\sum_{j_2}\sum_{t}QF'_{j_2} * V_{m,i,j_2,t} \quad (26)$$

4.3. Model Constraints

$$Z_{ij} \le Y_j \quad \forall i, j \quad (27)$$

$$\sum_{m}\sum_{i}V_{mijt} \le \tilde{C}_j \quad \forall j, t \quad (28)$$

$$\sum_{m}\sum_{j}V_{mijt} \le \tilde{S}_{it} \quad \forall i, t \quad (29)$$

$$\delta_{ij} * Z_{ij} \le H \quad \forall i, j \quad (30)$$

$$Z'_{jk}\left(\frac{\delta'_{jk}}{SP}\right) \le TI \quad \forall j, k \quad (31)$$

$$\sum_{i}V_{mijt} = \sum_{k}V'_{mjkt} \quad \forall m, j, t \quad (32)$$

$$\varphi_m * \delta_{pm} * \sum_{j}V'_{mjkt} = Q_{pmkt} \quad \forall p, m, k, t \quad (33)$$

$$\sum_{p}PT_p * G_{pkt} \le \widetilde{OT}_{kt} \quad \forall k, t \quad (34)$$

$$-\Psi \le X_{it} - X_{i't} \le \Psi \quad \forall i, i', t \quad (35)$$

$$Z_{ij}, Z'_{jk} \in \{0,1\} \quad \forall i, j, k \quad (36)$$

4.4. Robust Framework

In this section, we utilize the robust optimization approach described in Section 3.2 to account for the uncertain environment. The robust counterpart of the deterministic programming model can be formulated as follows:

$$\text{Min } \xi \quad (37)$$

$$\sum_{j_1}\left(E^E_{j_1} + F^F_{j_1} + LA^{LA}_{j_1}\right) + \sum_{j_2}\left(E^E_{j_2} + F^F_{j_2}\right) + \sum_{m}\sum_{i}\sum_{j}\sum_{t}P^{OC}_m * V_{mijt} + \sum_{m,j,k,t}P^{TC}_{j,k} * V'_{mjkt} + \cdots + \lambda^0 * \Gamma^0 \le \xi \quad (38)$$

5. Case Description

[National / Regional Organization] has been working since [Year]. According to statistics, [Organization] includes [N1] [facilities] and [N2] regional centers for testing and processing which are located in [N3] provinces of the country [**?**]. To illustrate the usefulness of the proposed model, this paper investigates a real case study of [City] which is the capital of [Region] and the [Nth] most populated city of [Country] having a population of about [X] million.

[City] consists of [N] districts, dispersed as shown in Fig. **??**, and plays a critical role in the provision of [product] for the country. Based on the above descriptions, the established [facilities] are in charge of collecting [product] from the [source] points. The properties of each district are presented in Table **??**. The properties of each candidate [facility] and each hospital are presented in Tables **??** and **??**, respectively.

With respect to [relevant standard], [product] demand can be estimated based on the patients' population or according to each hospital's bed. Transportation cost between each pair of facilities is proportional to the traveling distance, estimated to be [X]$ per kilometer.

6. Results and Discussions

In this section, the first phase is implemented by applying software [Name] for the purpose of obtaining the importance degree of each candidate local and mobile location.

Step 1. Calculating the weights of the DMs: in this part, the importance degree of the DMs is computed according to Section 3.1.2. Using Eqs. (**??**) and (**??**), the weights of the [N] DMs are obtained as follows: $\Gamma_1 = 0.161$, $\Gamma_2 = 0.173$, $\Gamma_3 = 0.138$, $\Gamma_4 = 0.166$, $\Gamma_5 = 0.194$, and $\Gamma_6 = 0.168$.

Step 2. Calculating the weights of the attributes: this part obtains the importance degree of each attribute based on model (**??**). According to the DMs' preference-intervals for each attribute, say $I = \{0.1 \le \omega_1 \le 0.2,\ 0.1 \le \omega_2 \le$

$0.3,\ldots\}$, the vector of attributes weights are computed as: $\omega = \{0.2, 0.1, 0.2, 0.3, 0.1, 0.1\}$.

Step 3. Identifying the candidate locations: in this part, the most qualified candidate locations are identified and ranked based on the obtained weights.

In continue, the computational experiments are carried out through CPLEX solver by utilizing software GAMS on a PC with Intel Core i7, CPU 2.4 GHz and 16 GB of RAM. As shown in Table **??**, the results obtained by robust and deterministic models are presented under the variability of uncertain parameters and conservatism level. For this purpose, we first set the values of the compensation coefficient (Υ) to 0.4 and the relative importance of the objective functions ($\vartheta_1, \vartheta_2, \vartheta_3$) to 0.6, 0.2, and 0.2, respectively.

A general expectation is that the network cost of the robust approach under parameters variability is higher than that of the deterministic mode since it imposes additional costs of handling uncertainty and making a stable network. By increasing the level of uncertainty, the cost increase becomes even more tangible. Furthermore, the influence of utilizing robust approach on computing time is negligible comparing to the deterministic attitude.

6.1. The Supply Chain Design Under Different Variability Levels of Uncertain Parameters

The impact of uncertainty levels on optimal structure of the supply chain provided by robust and deterministic approaches is mapped and compared in Fig. **??**. A general understanding is that in addition to the existing local [facilities], by increasing the variability from 10% to 50%, a slight increase in the number of mobile [facilities] is justified to cover the realized perturbation.

6.2. The Model Performance Under Different Levels of Compensation Coefficients (Γ)

In this part, the impact of varying Γ on the performance of the model is evaluated. As the value of Γ increases the model forces the membership functions to get closer to each other while on the other hand, the lower levels of Γ give the importance to the objective with higher weight.

6.3. The Model Performance Under Different Levels of Conservatism and Data Variability

This section aims to analyze the behavior of the objective function of cost while adopting different risk-averse attitudes and variability levels. The analysis is presented in three parts: (a) Sensitivity of the model to an uncertain production cost, (b) Sensitivity of the model to an uncertain capacity of a [facility], and (c) Sensitivity of the model to uncertain demand.

7. Discussions and Conclusions

Designing [domain] supply chain networks is tainted with a great degree of intricacies due to properties including rather irregular supply and demand, wastage and shortage costs as well as [domain]-related considerations (e.g., perishability of [product]), which are not easily tractable. With this significance in mind, this paper proposes a novel 3P-MAMOR model to formulate the importance of the factors above to best match the real world conditions. The proposed model aims to consider both cost-oriented and quality-oriented approaches by minimizing the overall network cost, preserving as much as possible the [product freshness], and seeking the most qualified locations for [facilities]. This paper also applies a real case study to accommodate the real world application of the proposed 3P-MAMOR model.

According to the results obtained from the solution, it seems imperative to make a tradeoff between cost-oriented and quality-oriented approaches. Qualitative factors like availability, market environment, and transportation conditions have significant effect on finding the preferable locations for collection facilities. Moreover, the robust approach has dominated the deterministic model in terms of the mean and SD of the value of the total network cost.

Future research can be directed in investigating the problem while using other MADM techniques such as fuzzy AHP or fuzzy TOPSIS to evaluate the critical factors involved. The interested practitioners can also develop the hospital level by accounting for compatibility procedures. Moreover, the mathematical modelling in realm of [donation system] could be developed by considering age-based variety of [products].

Table 2. The properties of each district.

District	Population	Rate (%)	Supply	($\mathbf{LONG}_i$, $\mathbf{LAT}_i$)
1	212,206	14	38	(38.077, 46.311)
2	169,047	11	29	(38.063, 46.287)
3	243,400	16	42	(38.067, 46.255)
4	316,126	21	55	(38.076, 46.253)
5	92,274	6	16	(38.048, 46.366)
6	94,897	6	16	(38.086, 46.232)
7	143,460	10	27	(38.043, 46.191)
8	28,700	2	6	(38.081, 46.298)
9	324	0	1	(38.014, 46.423)
10	194,564	13	34	(38.112, 46.290)

Table 1. Nomenclature of the investigated papers' characteristics.

Hierarchical level		**Modelling approach**		**Performance measure**	
Collection	CO	Simulation	S	Cost	C
Production	PR	Statistical analysis	SA	Distance	D
Inventory	IN	(Stochastic) DP	(S)DP	Shortage	SH
Distribution	DI	Integer Programming	IP	Outdates	O
Integrated	INT	(M)INLP	(M)INLP	Freshness	F
Planning horizon		**Type of blood product**		**Uncertainty approach**	
Strategic	STR	Whole blood	WB	Fuzzy	FZ
Tactical	TAC	Platelet	PLT	Stochastic	ST
Operational	OPR	Plasma	PLS	Robust	RO
Solution method		**Objective function**			
Exact	EX	Single-objective	SO		
(Meta)heuristic	ME	Multi-objective	MO		

Table 3. The properties of each candidate local and mobile [facility].

Row	**Facility**	**Type**	**(LONG$_i$, LAT$_i$)**
1	[Facility Name 1]	Candidate local	(38.073, 46.252)
2	[Facility Name 2]	Candidate local	(38.073, 46.303)
3	[Facility Name 3]	Candidate local	(38.066, 46.311)
4	[Facility Name 4]	Candidate local	(38.076, 46.288)
5	[Facility Name 5]	Candidate local	(38.065, 46.271)
6	[Facility Name 6]	Candidate local	(38.059, 46.348)
1	[Mobile 1]	Candidate mobile	(38.055, 46.278)
2	[Mobile 2]	Candidate mobile	(38.061, 46.333)
3	[Mobile 3]	Candidate mobile	(38.084, 46.265)
4	[Mobile 4]	Candidate mobile	(38.063, 46.303)
5	[Mobile 5]	Candidate mobile	(38.091, 46.264)

Table 4. The properties of each demand zone.

Row	**Hospital**	**Beds**	**(LONG$_i$, LAT$_i$)**
1	[Hospital 1]	599	(38.108, 46.302)
2	[Hospital 2]	67	(38.093, 46.279)
3	[Hospital 3]	85	(38.076, 46.259)
4	[Hospital 4]	154	(38.077, 46.279)
5	[Hospital 5]	307	(38.082, 46.306)
6	[Hospital 6]	168	(38.080, 46.306)
7	[Hospital 7]	56	(38.069, 46.297)
8	[Hospital 8]	37	(38.068, 46.297)
9	[Hospital 9]	124	(38.060, 46.282)
10	[Hospital 10]	93	(38.078, 46.318)

Table 6. The model performance under different levels of compensation coefficients (Γ).

Prob.	Γ	$\vartheta_1, \vartheta_2, \vartheta_3$	Z_1	Z_2	Z_3	μ_1	μ_2	μ_3
1	0.0		331,717	410	511	0.995	0.996	0.355
2	0.1		331,698	413	511	0.995	0.996	0.355
3	0.2		421,223	418	592	0.944	0.996	0.468
4	0.3		708,844	348	812	0.781	0.996	0.776
5	0.4	$\vartheta_1 = 0.7$	709,501	343	812	0.780	0.996	0.776
6	0.5	$\vartheta_2 = 0.2$	706,425	418	812	0.780	0.996	0.776
7	0.6	$\vartheta_3 = 0.1$	706,425	418	812	0.781	0.996	0.776
8	0.7		706,425	418	812	0.781	0.996	0.776
9	0.8		706,425	418	812	0.781	0.996	0.776
10	0.9		706,322	440	812	0.781	0.995	0.776
11	1.0		716,357	11884	812	0.776	0.874	0.776

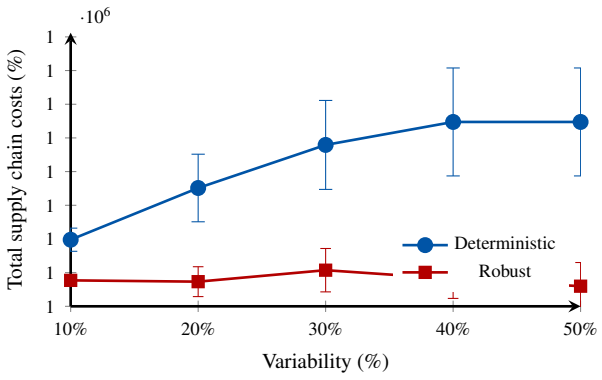


Figure 3. The mean and standard deviation of the first objective function value (Conservatism Level 90%).

Table 5. Summary of results under uncertain parameters and deterministic condition ($\Gamma = 0.4$, $\vartheta_1 = 0.6$, $\vartheta_2 = \vartheta_3 = 0.2$).

Var.	Deterministic Z_1	Robust Z_1 — Conservatism level				Det. Z_2	Det. time (s)
		60%	70%	80%	90%		
0%	682,918	—				688	72
10%	698,008	683,849	690,235	671,599	661,632	589	—
20%	710,753	709,518	697,482	678,357	658,549	491	190
30%	708,092	695,032	739,653	708,092	728,294	305	282
40%	761,267	744,372	742,249	663,780	767,346	586	231
50%	775,397	794,836	786,078	680,403	773,581	345	206

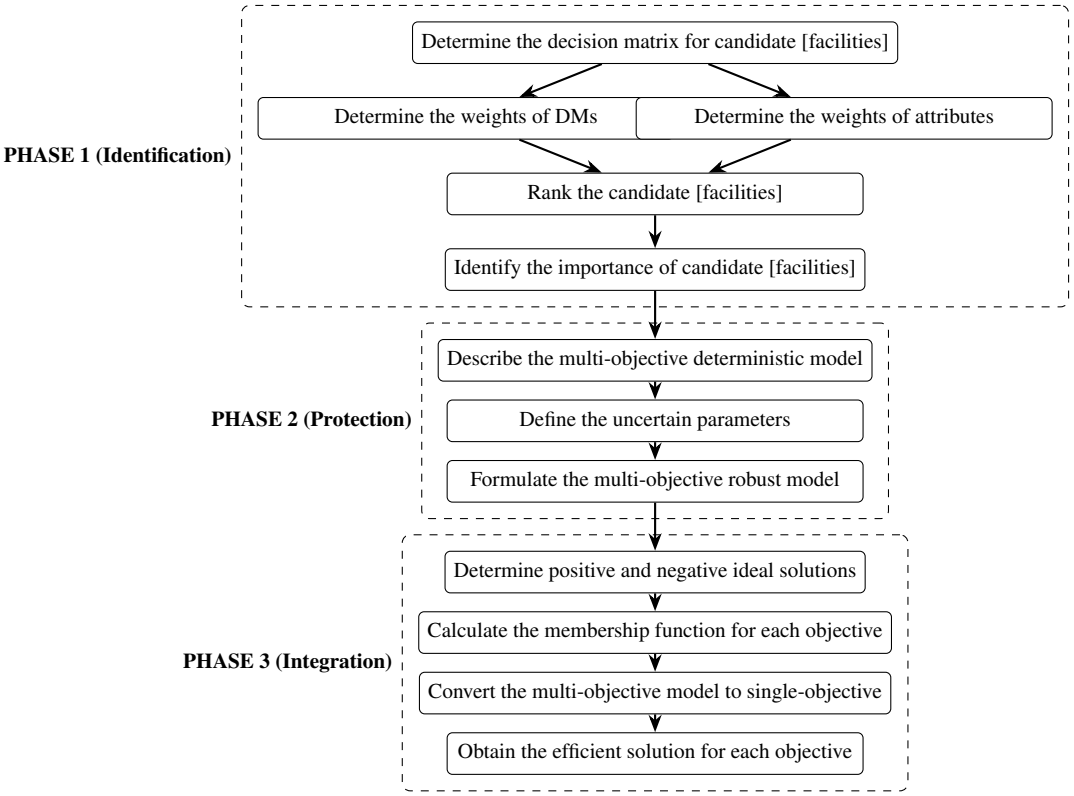


Figure 1. Visual representation of the 3P-MAMOR modelling approach.

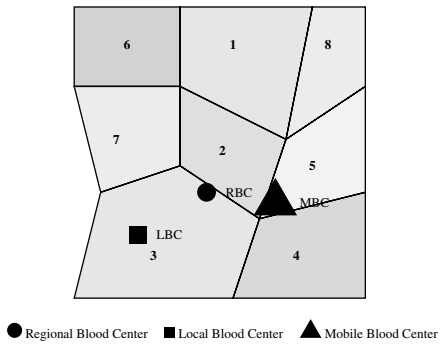


Figure 4. Geographic dispersion of districts of [City].

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cie.2019.XX.XXX.

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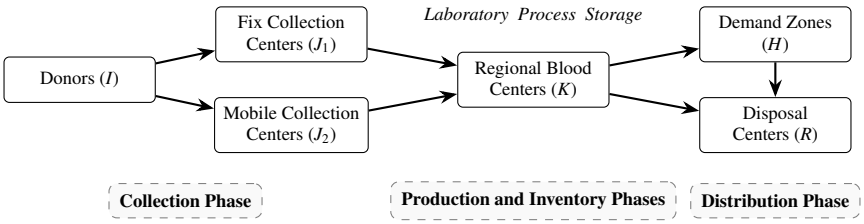


Figure 2. The concerned [domain] supply chain overview.

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